

Python Modules Cheat Sheet

1. Modules (Basic Concept)

Modules are reusable Python files containing functions, variables, and classes.

2. math Module

Used for mathematical operations.

`sqrt(x)`, `pow(x,y)`, `ceil(x)`, `floor(x)`, `log(x)`, `sin(x)`, `cos(x)`, `tan(x)`

Constants: `pi`, `e`

3. random Module

Generates random values.

`random()`, `randint(a,b)`, `uniform(a,b)`, `choice(list)`, `shuffle(list)`

4. string Module

`ascii_letters`, `digits`, `punctuation`

Used for password generation and text processing.

5. datetime Module

Handles date and time.

`now()`, `today()`, `strftime()`, `strptime()`, `timedelta`

Used for formatting, calculations, and timestamps.

6. join() Function

Combines iterable items into a string efficiently.

Example: `".join(['A','B','C'])` -> `'ABC'`